

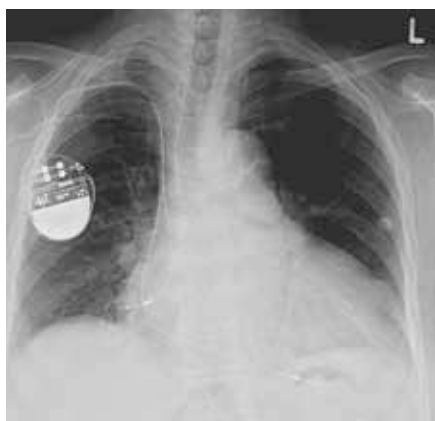
haven't already patched them, which means the attacker now has a path to your system provided you have that particular software installed.

It doesn't matter whether you're on Windows, Linux or OS X, every platform is vulnerable. Using Metasploit and Kali Linux the attacker can create a backdoor in as little as four steps. This is just a backup in case the Zero-Day vulnerability gets patched, there will always be another way into your computers. The attacker can then simply monitor the incoming and outgoing traffic using Wireshark. This versatile tool is capable of analysing and generating detailed reports which means your entire online activity is now in the grasp of someone who wishes to cause you harm.

Internal implants, pace makers

If you haven't already started, now's a great time to start wearing a tin foil hat and a body suit lined with lead plates because what you're going to read next will push you over the edge (maybe literally).

Advancements in medical science have not only vastly improved the standard of living for millions of people worldwide, they're also opening the doors to hackers,



Pacemakers are hackable too!

potentially giving them access to your very life. Pacemakers and drug pumps have featured wireless connectivity since 2004 (yep, that's 10 years), ostensibly, to give doctors direct access to the device and to enable them to monitor the state of your health.

Unfortunately, in 2008, a team of researchers and hackers managed to gain control of one such device in a laboratory environment and managed to deliver jolts of electricity and also to shutdown the



Why bother hacking when these equipment have simple hardcoded passwords like '1234'?

device. Of course, they required about \$30,000 of lab equipment and a whole team of experts, but this was in 2008. A lot can happen in six years, and what about 10 years hence?

If this little tid-bit of information wasn't enough to send a chill up your spine (with that spinal implant, which might actually be a real chill), hackers have demonstrated their ability to hack insulin pumps from certain manufacturers. This gave the hackers complete control over the dosage of insulin in every machine in a 300m radius and the consequences can be quite chilling.

But this is still not the scary part. A 2014 article in *Wired* showcased an investigative story where a security analyst inspected a hospital, only to come back shell-shocked at the utter lack of any form of security in crucial equipment. Everything from drug pumps to X-ray machines to defibrillators and myriad other equipment was completely compromised and if even that wasn't bad enough, it turned out that passwords like "1234" were hard-coded into the devices. Attackers with the right know-how can literally manipulate medical records, remotely access X-ray data and other such information.

Some of these implants don't even need to be hacked. They are often vulnerable to microwave bursts and that too on frequencies commonly used in Wi-Fi and cordless phones. Microwave Hyperthermic Lipolysis is a procedure by which fatty tissue is liquefied using a wireless RF antenna set to 5.8GHz. This is a very common frequency band that is also used in cordless phones. Even DBS (Deep Brain Stimulation) implants are vulnerable to bursts of microwave energy.

Pwned

It's shockingly easy to get hacked to death, in the right (or wrong) circumstances and the worst part is, it'll be incredibly difficult to prove. Imagine sitting at your

home and having your PC suddenly catch fire (<http://dgit.in/1nGFt2t>), or your printer start spewing out burnt pages (<http://dgit.in/10tMcTd>, in HP's defence, there's not



A typical IMSI catcher

much meat on that proposition), how would you be able to separate coincidence from malicious intent? We're living in a world where anything and everything is connected to the Wild Wild Web, and you're at the very heart of it. The Internet has you, and you're nothing but a mere pawn. **d**